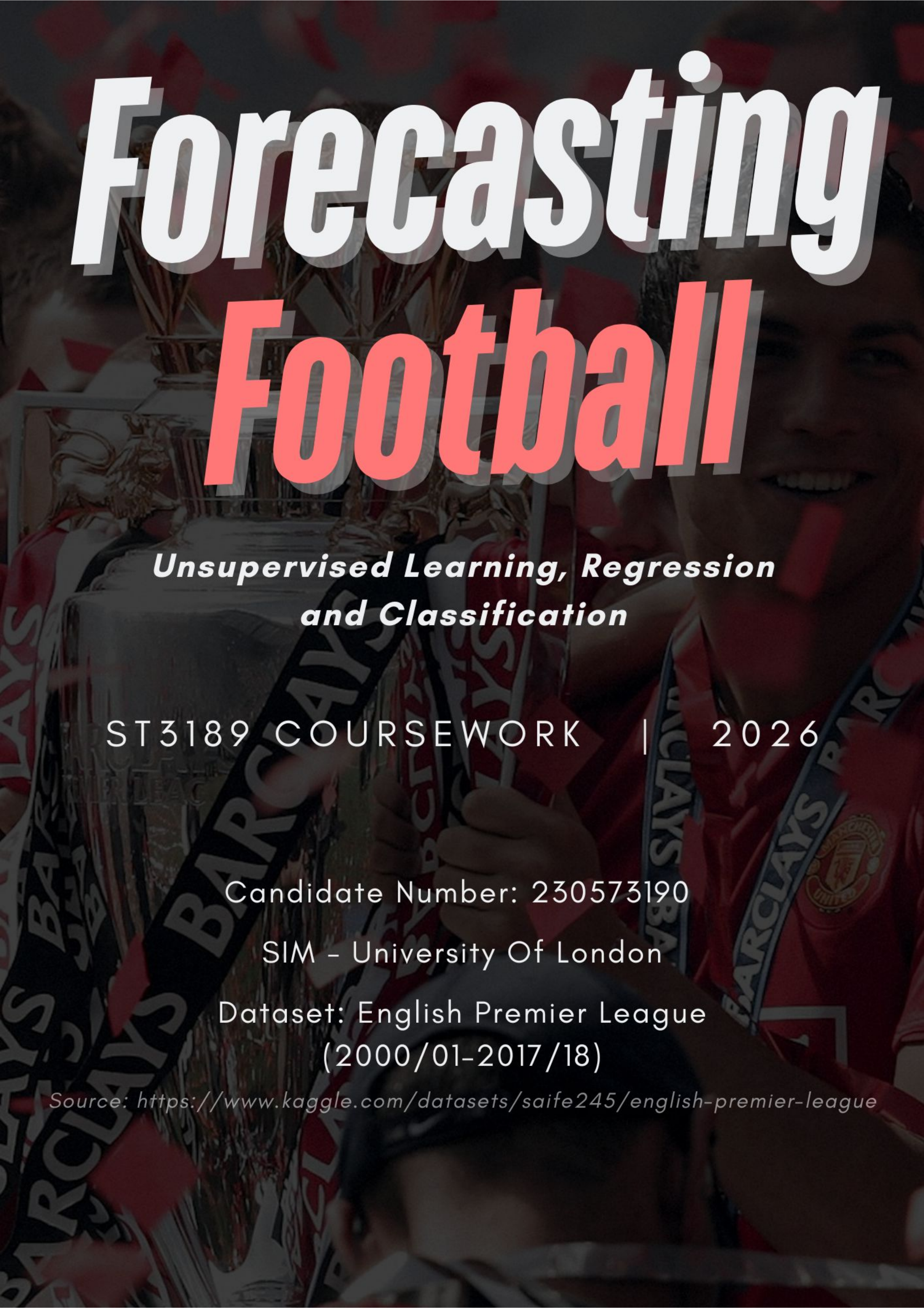

Forecasting Football

Unsupervised Learning, Regression and Classification

ST3189 COURSEWORK | 2026

Candidate Number: 230573190

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Dataset: English Premier League (2000/01-2017/18)

Source: https://www.kaggle.com/datasets/saife245/english-premier-league

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Executive Summary

This report applies machine learning to 6,840 English Premier League (EPL) match records (2000/01–2017/18) across three tasks.

Unsupervised learning: PCA identifies two dominant variance dimensions: match competitiveness (PC1, 31.6%) and overall match quality (PC2, 24.1%). K-means clustering (K=3) reveals three interpretable match contexts: *Balanced* (home win rate 46%), *Home Favoured* (65%), and *Away Favoured* (29%).

Regression: OLS and Ridge both achieve a test RMSE of **1.229 goals**, with home goal difference per game (HTGD) the dominant predictor.

Classification: Logistic Ridge achieves the best test AUC of **0.695** (accuracy 64.9%), consistently outperforming KNN and Random Forest. Across both supervised tasks, linear models match or exceed tree-based methods, indicating a predominantly linear predictive structure in pre-match EPL statistics.

1. Introduction

The English Premier League (EPL) is the world's most commercially significant football competition. Pre-match prediction using publicly available team statistics represents a practically useful modelling exercise. Prior work has demonstrated machine learning's viability: *Raju et al. (2020)* achieved 68.55% accuracy with Random Forest over 10 EPL seasons; *Choi, Foo and Chua (2023)* reported 77.43% binary accuracy combining player ratings with form data; *Prasetio and Damanik (2016)* established logistic regression as a competitive baseline. A consistent finding across this literature is that ***recent form*** and ***goal-based strength metrics*** are the most informative pre-match predictors *(Rodrigues and Pinto, 2022).*

This study addresses three research questions:

- RQ1 (Unsupervised): Do pre-match team statistics reveal distinct and interpretable match archetypes?
- RQ2 (Regression): Which features best predict home goals scored, and how accurately?
- RQ3 (Classification): Can pre-match data classify whether the home team wins, and which model performs best?

All analyses use only pre-match cumulative statistics, ensuring no leakage from within-match events.

2. Data Overview

The dataset is sourced from Kaggle (https://www.kaggle.com/datasets/saife245/english-premier-league). It contains 6,840 EPL matches across 18 seasons, with 13 pre-match predictor variables described in Table 1 below:

Table 1: Description of the 13 Pre-match Predictor Variables

Variable	Description
HTP	Home team points per game (cumulative to this match)
ATP	Away team points per game (cumulative to this match)
HTGD	Home team goal difference per game (cumulative)
ATGD	Away team goal difference per game (cumulative)
HTFormPts	Home team points from last 5 matches
ATFormPts	Away team points from last 5 matches
DiffPts	HTP minus ATP -> overall relative strength
DiffFormPts	HTFormPts minus ATFormPts -> relative recent form
HTWinStreak3 / HTLossStreak3	Home team on 3+ win / loss streak (1=yes, 0=no)
ATWinStreak3 / ATLossStreak3	Away team on 3+ win / loss streak (1=yes, 0=no)
MW	Match week within the season (1–38)

HT = Home Team; AT = Away Team. All statistics computed from matches played prior to the current fixture.

The **regression target** is FTHG (Full Time Home Goals, mean 1.53). The **classification target** is a binary indicator of whether the home team won (FTR = 'H'), with a home win rate of 46.4% across the dataset; a meaningful base rate reflecting the well-documented *home advantage effect (Leite, 2017)*.

A **temporal split** is applied: the first 5,472 matches (80%, seasons 2000/01–2014/15) form the training set; the remaining 1,368 matches (20%, seasons 2015/16–2017/18) form the test set. All features are standardised within the training set; the same scaling is applied to the test set.

Figure 1 summarises key distributional patterns. Home goals (graph a) and away goals (graph b) both exhibits right-skewed, count distributions with means of 1.53 and 1.13 respectively, confirming the *home scoring advantage*. Graphs (c) and (d) show that both home points per game (HTP) and home goal difference per game (HTGD) are clearly **higher** in home-win matches, validating these as candidate predictors.

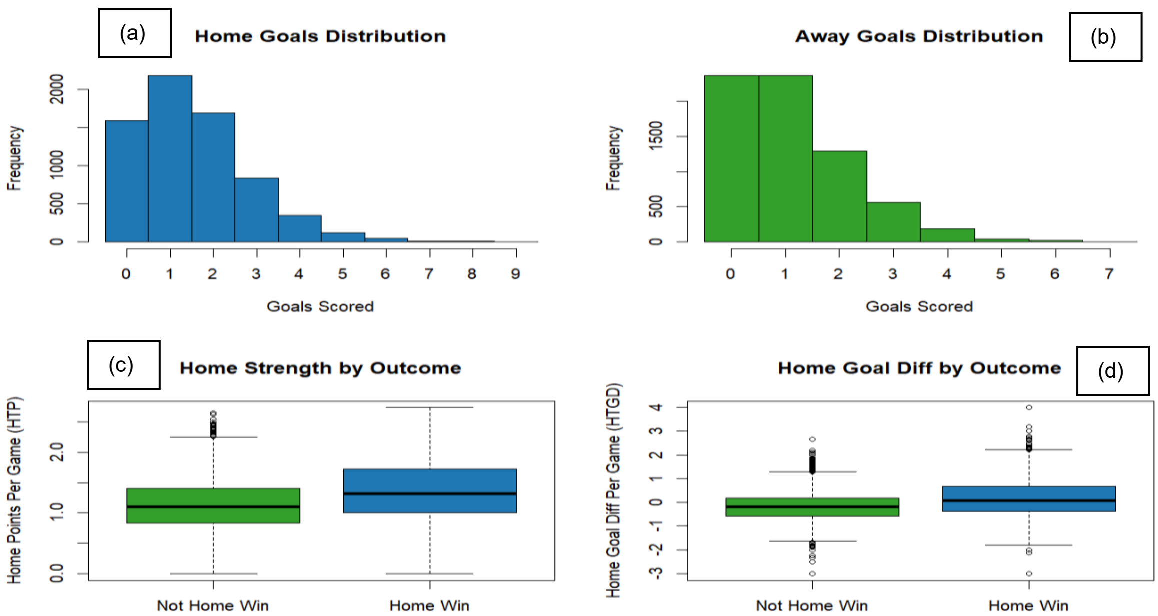

Figure 1: Exploratory Data Analysis. (a) Home goals distribution. (b) Away goals distribution. (c)–(d) Home strength metrics by match outcome.

3. Unsupervised Learning - PCA and Clustering

We applied unsupervised learning to the 13 pre-match features to (i) reduce dimensionality for visualisation and (ii) identify interpretable match archetypes that contextualise the supervised analyses.

3.1 Principal Component Analysis

PCA is applied to the standardised 13-feature training data. The scree plot and cumulative variance chart (Figure 2) reveal a clear elbow after PC2. The first two components jointly explain **55.7%** of variance (PC1: 31.6%; PC2: 24.1%), rising to 78.6% by PC5.

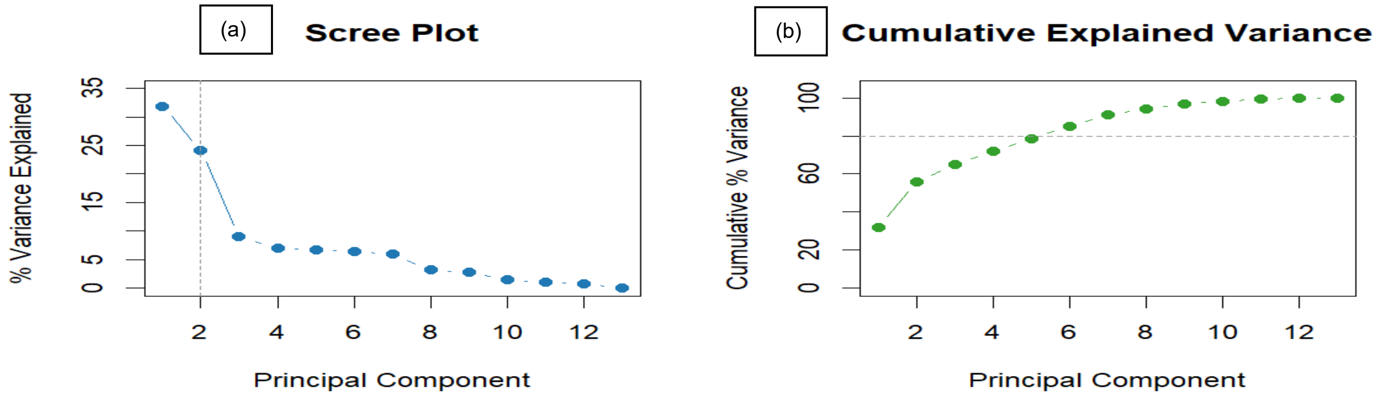

Figure 2: PCA results. (a) Scree plot showing the elbow after PC2. (b) Cumulative explained variance; dashed line at 80% threshold.

PC1 (Match Competitiveness): Led by DiffPts (0.472), DiffFormPts (0.418), HTGD (0.332), and HTP (0.320). A positive PC1 score indicates a home-favoured fixture while negative indicates an away-favoured one.

PC2 (Overall Match Quality): Led by ATP (0.428), ATFormPts (0.418), HTP (0.387), and HTFormPts (0.383), all loading in the same direction. This component captures matches where *both* teams perform well simultaneously & distinguishes high-quality contests (both strong teams) from low-quality ones (both weak), independent of which team is favoured.

These two components provide clean, interpretable two-dimensional representations of match context, directly informing the clustering analysis below.

3.2 K-means Clustering

K-means is applied to the standardised training data. The elbow method (Figure 3a) shows the steepest reduction at K=3, with minimal additional gain at K=4 and beyond. Three clusters are therefore selected. Each run used 25 random initialisations to avoid local optima.

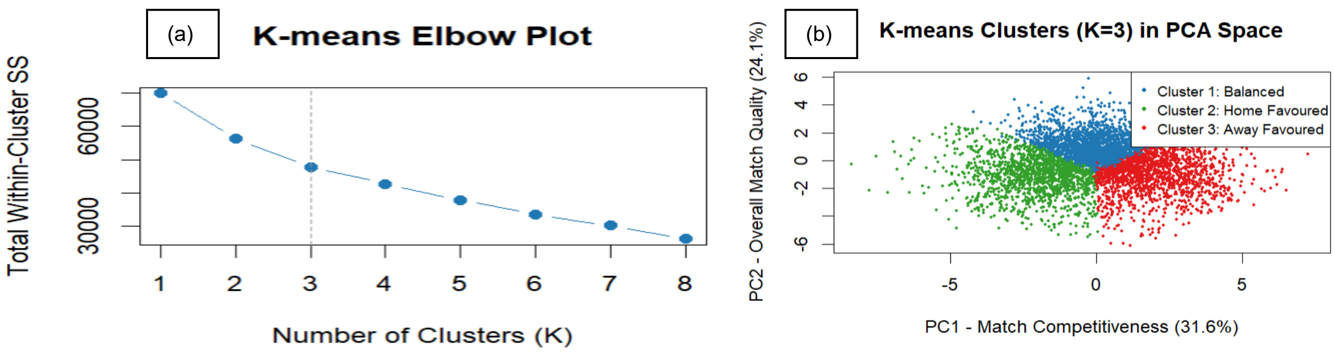

Figure 3: K-means clustering. (a) Elbow method; sharpest bend at K=3. (b) Clusters projected onto PC1–PC2 space, showing clear separation along the match competitiveness axis.

Figure 3(b) confirms that the three clusters separate cleanly along PC1 (match competitiveness), consistent with the PCA findings. Table 2 profiles each cluster:

Table 2: K-means Cluster Profiles (K=3, Training Set)

Cluster	n	DiffPts	DiffFormPts	HTGD	HW Rate	Avg HG
Cluster 1: Balanced	2,285	0.00	0.00	-0.36	46.1%	1.49
Cluster 2: Home Favoured	1,618	0.64	0.30	0.73	64.5%	1.95
Cluster 3: Away Favoured	1,569	-0.73	-0.37	-0.26	29.1%	1.14

HW Rate = Home Win Rate; Avg HG = Average Home Goals. DiffPts and DiffFormPts in original units.

Cluster 1 (Balanced) (n=2,285, 41.7%): Near-equal teams; home win rate 46.1%, average home goals 1.49.

Cluster 2 (Home Favoured) (n=1,618, 29.6%): Home team holds a clear season and form advantage; home win rate 64.5%, average home goals 1.95.

Cluster 3 (Away Favoured) (n=1,569, 28.7%): Away team dominates on season points and goal creation; home win rate 29.1%, average home goals 1.14. The 0.81-goal spread in average home goals across clusters demonstrates strong downstream relevance for the regression task.

3.3 Hierarchical Clustering

For robustness, hierarchical clustering with complete linkage is also applied to a random subsample of 300 training observations. Cutting the dendrogram at height h=4.8 yields three clusters consistent in character with the K-means solution, providing confirmatory evidence that the three-cluster structure is stable and not an artefact of initialisation. The dendrogram also reveals a natural distance of approximately 4.8 units separating the three groups, with no ambiguous intermediate structure suggesting K=2 or K=4 would be more appropriate.

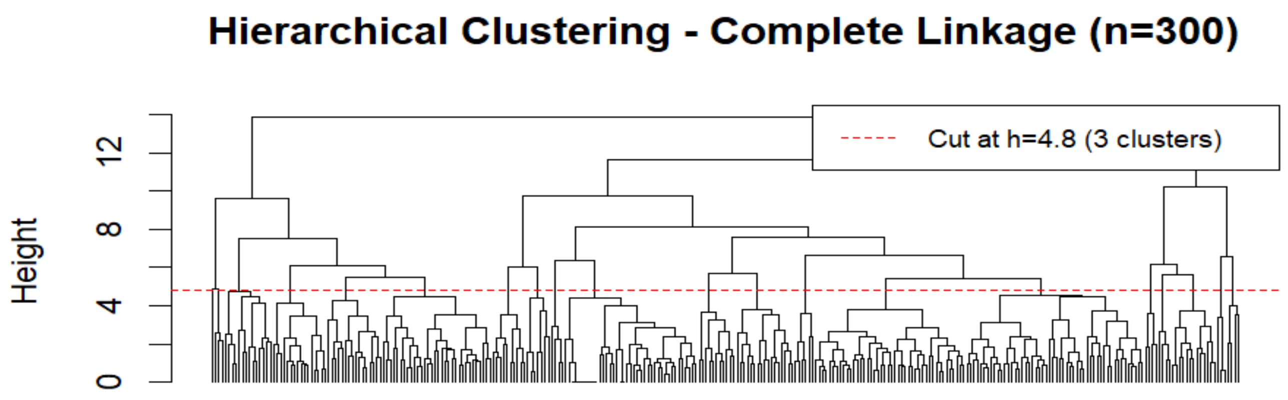

Figure 4: Hierarchical dendrogram (complete linkage, n=300 subsample). Dashed line at h=4.8 yields three clusters matching the K-means solution.

4. Regression - Predicting Home Goals

Four models predict FTHG from the 13 pre-match features using a scaling pipeline to prevent test-set leakage. Model selection is based on 5-fold cross-validated RMSE.

4.1 Model Comparison

Table 3: Regression Model Performance (Test Set, n=1,368)

Model	CV-RMSE	Test-RMSE	Notes
Null Baseline (mean prediction)	-	1.297	Reference lower bound
OLS Linear Regression	**1.240**	**1.229**	Best (tied with Ridge)
Ridge Regression	**1.240**	**1.229**	Matches OLS (low collinearity)
Lasso Regression	1.245	1.234	Retains 3 of 13 features
Random Forest	1.297	1.268	Non-linear; no improvement

Green rows = best models. RMSE = Root Mean Squared Error; lower is better.

- OLS and Ridge regression are jointly best, both achieving a test RMSE of **1.229 goals**, compared to the null baseline RMSE of 1.297. This represents a 5.2% improvement in predictive error. The near-identical performance of OLS and Ridge is informative: it indicates that *multicollinearity is not a material problem* in this feature set, so the L2 penalty has nothing additional to offer over OLS.

- Lasso (RMSE = 1.234) selects only **three features** at its cross-validated optimal penalty: HTGD (coeff = 0.162), DiffPts (0.134), and ATGD (-0.096). The fact that 10 of 13 features are shrunk exactly to zero with a predictive performance that is barely degraded; confirms that a simple linear model is sufficient and that most of the predictive signal is captured by goal-based metrics.

- Random Forest performs the worst (RMSE = 1.268), suggesting that the relationship between pre-match features and home goals is *approximately linear*, where non-linear interactions do not provide additional signal beyond what the linear models capture. This aligns with the broader sports analytics' literature finding that simple linear models often match or outperform complex ensemble methods when features are already well-engineered (Rodrigues and Pinto, 2022).

4.2 Feature Importance

We observe that the Ridge coefficient profile identifies a clear hierarchy.

HTGD (coeff = +0.258) is the strongest positive predictor: a superior home goal difference per game directly increases expected home scoring.

ATGD (coeff = -0.180) is the strongest negative: a stronger away defence suppresses home goals.

DiffFormPts (coeff = -0.171) indicates that when the away team holds a recent form advantage, home scoring falls. Streak indicators (HTWinStreak3, HTLossStreak3) carry near-zero coefficients, consistent with Lasso excluding them entirely. This aligns with *Choi et al. (2023)*, who similarly found goal difference and form to be the most informative pre-match EPL features.

5. Classification - Predicting Home Win

Five models predict whether the home team wins (HomeWin vs NotHomeWin) from the same 13 features. AUC is the primary metric (threshold-independent, robust to the 46/54 class imbalance). All models use probability outputs to enable ROC analysis.

5.1 Model Comparison and ROC Analysis

Table 4: Classification Model Performance (Test Set, n=1,368)

Model	CV-AUC	Test-AUC	Test-Acc	Notes
Logistic Regression (OLS)	0.679	0.695	64.8%	Baseline
Logistic Regression (Ridge)	0.679	0.695	64.9%	Best overall
Logistic Regression (Lasso)	0.679	0.695	64.8%	Comparable to Ridge
K-Nearest Neighbours (k=15)	0.630	0.652	59.4%	Non-linear baseline
Random Forest	0.634	0.671	63.3%	Ensemble; below logistic

Green row = best model. AUC = Area Under ROC Curve; Acc = Accuracy. All models use predict_type = "prob".

All three logistic variants achieve identical CV-AUC (0.679) and test AUC (0.695), with Ridge marginally best on accuracy (64.9%). KNN (test AUC 0.652) and Random Forest (0.671) both underperform, confirming that the home-win decision boundary is approximately linear in this feature space.

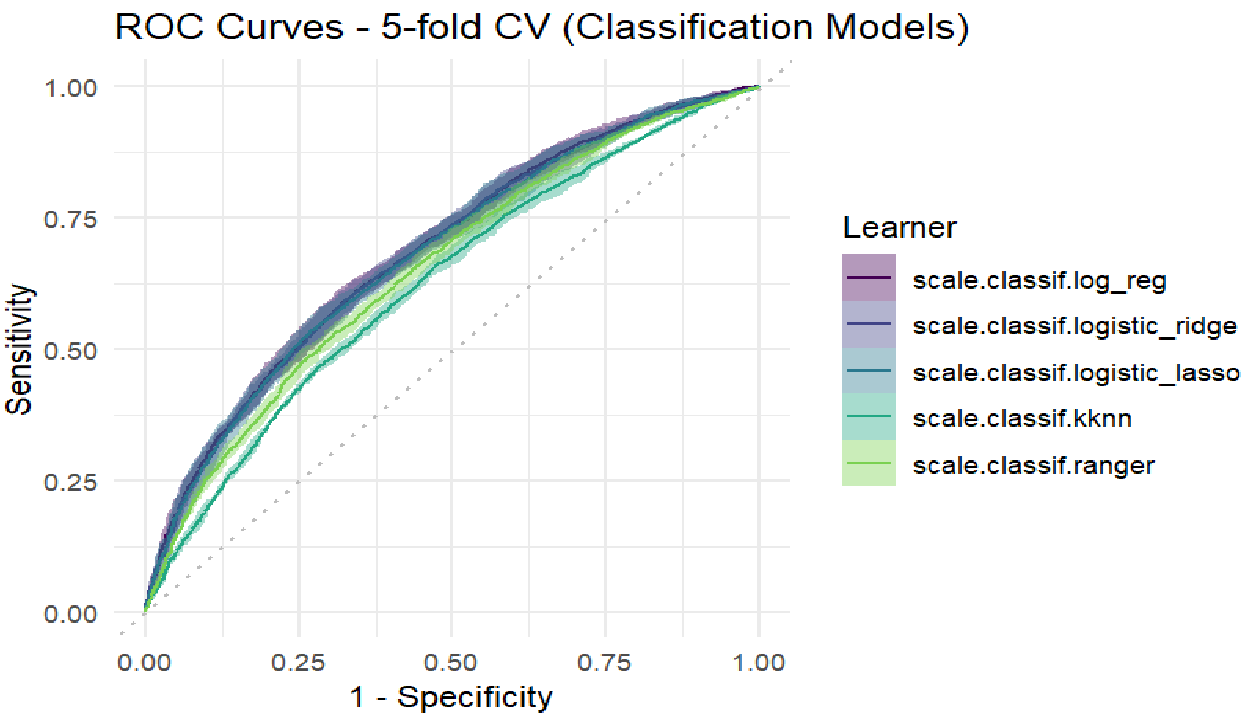

Figure 5: ROC curves from 5-fold cross-validation. The three logistic models closely overlap near the top of the plot, with Random Forest and KNN trailing below them. The dashed diagonal represents random classification (AUC = 0.500).

Figure 5 makes the separation visible: the three logistic curves cluster near the top-left (high sensitivity, low false-positive rate), while KNN and Random Forest fall below them across all thresholds. The AUC gap between the best logistic model (0.679 CV) and KNN (0.630 CV) is practically meaningful, representing a 7.3% relative improvement.

5.2 Best Model: Logistic Ridge

Table 5: Confusion Matrix; Logistic Ridge (Test Set)

	Predicted: NOT Home Win	Predicted: Home Win
Actual: NOT Home Win	540 (TN)	205 (FP)
Actual: Home Win	275 (FN)	348 (TP)

Accuracy = 64.9%; Sensitivity = 55.9%; Specificity = 72.5%. TN/TP = green (correct); FP/FN = red (errors).

The model correctly identifies 72.5% of non-home-wins (specificity) but only 55.9% of home wins (sensitivity). This asymmetry is expected: home victories carry higher inherent uncertainty, as the home advantage is a probabilistic tendency rather than a guarantee. Figure 6 plots the standardised coefficient magnitudes.

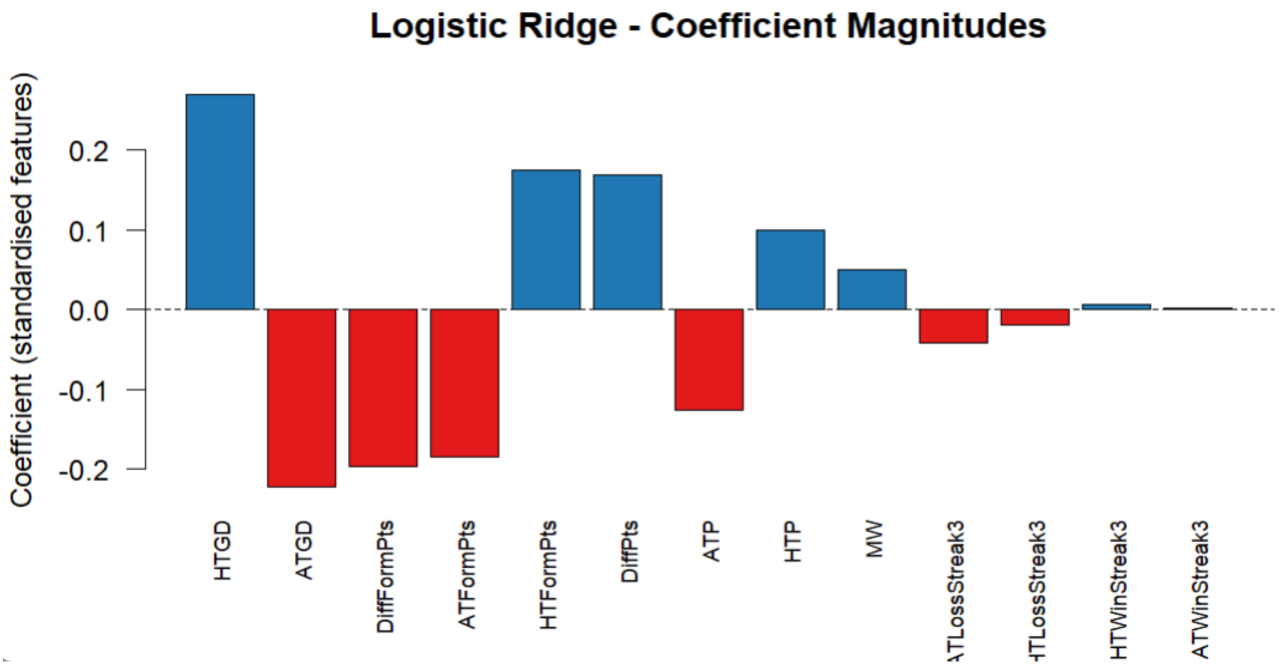

Figure 6: Logistic Ridge standardised coefficients. Blue bars increase home win probability; red bars decrease it. HTGD is the strongest positive predictor; DiffFormPts and ATGD are the strongest negative predictors.

HTGD (coeff = +0.324) is the single strongest predictor. **DiffFormPts** (coeff = -0.279) and **ATFormPts** (coeff = -0.237) reflect that an away team in good recent form substantially reduces home win probability. **DiffPts** (coeff = +0.158) confirms season-level dominance matters. Streak indicators contribute negligible coefficients, suggesting their value is already captured by the cumulative form and points-per-game variables.

6. Discussion and Conclusion

In a nutshell, the three analyses tell a coherent story about EPL match predictability using pre-match team statistics. The unsupervised analysis establishes that pre-match data is not informationally flat: three meaningful match contexts exist (Balanced, Home Favoured, Away Favoured) with dramatically different goal-scoring and win-rate profiles. This structural insight validates the use of the same feature set for the supervised tasks and provides context as to why certain matches are harder to predict than others.

Across both supervised tasks, **goal difference per game (HTGD, ATGD)** and **relative form (DiffFormPts, ATFormPts)** consistently emerge as the most informative predictors. This is consistent with the football analytics literature, providing a clear & actionable message: teams wishing to forecast performance should prioritise goal-based metrics over win/loss streaks or simple points totals. The convergence of linear models (OLS, Ridge, logistic) with near-identical performance, and their consistent outperformance of non-linear methods (KNN, Random Forest), indicates that the predictive structure in pre-match EPL data is predominantly linear, where non-linear interactions between features do not add explanatory value with this feature set.

Limitations. First, the pre-match framework excludes within-match statistics (shots, possession, corners) that contain substantial predictive information. Studies incorporating such features report higher accuracy, though they require real-time data. Second, team identity is not used as a feature, meaning fixed effects for historically dominant clubs (e.g. Manchester United, Arsenal) are not captured. Third, the model assumes stationarity across 18 seasons; changes in tactics, squad investment, and league competitive balance over time may reduce model applicability, especially in recent seasons as more EPL clubs undergo ownership takeovers that structurally change the directions of the football club. Lastly, the class boundary between HomeWin and NotHomeWin is inherently noisy, as football retains substantial irreducible uncertainty *(Pappalardo and Cintia, 2018)*.

Recommendations. For practitioners seeking pre-match predictions, the Logistic Ridge model is recommended: it is interpretable, computationally trivial, and achieves the best AUC in this study. Teams or analysts wishing to improve upon the 64.9% accuracy ceiling should (i) incorporate rolling shot-based metrics (xG proxies) using only pre-match public data, (ii) include team fixed effects to capture chronic strength differences, and (iii) explore time-aware cross-validation (e.g. forward-chaining) to account for the non-stationarity of the EPL across seasons.

Conclusion. This study demonstrates that machine learning applied to pre-match EPL team statistics produces interpretable and practically useful models for both goal prediction (RMSE = 1.229) and match outcome classification (AUC = 0.695). The three machine learning tasks; unsupervised clustering, regression, and classification are mutually reinforcing: unsupervised learning reveals the structural context that explains why match prediction is difficult, while the supervised analyses quantify the extent to which that structure can be exploited. Goal-based performance metrics, particularly home and away goal difference per game, provide the backbone for predictive power. These findings offer a principled, data-driven foundation for pre-match analysis in arguably the world's most closely scrutinised sporting competitions, better known as *“The Beautiful Game”.*

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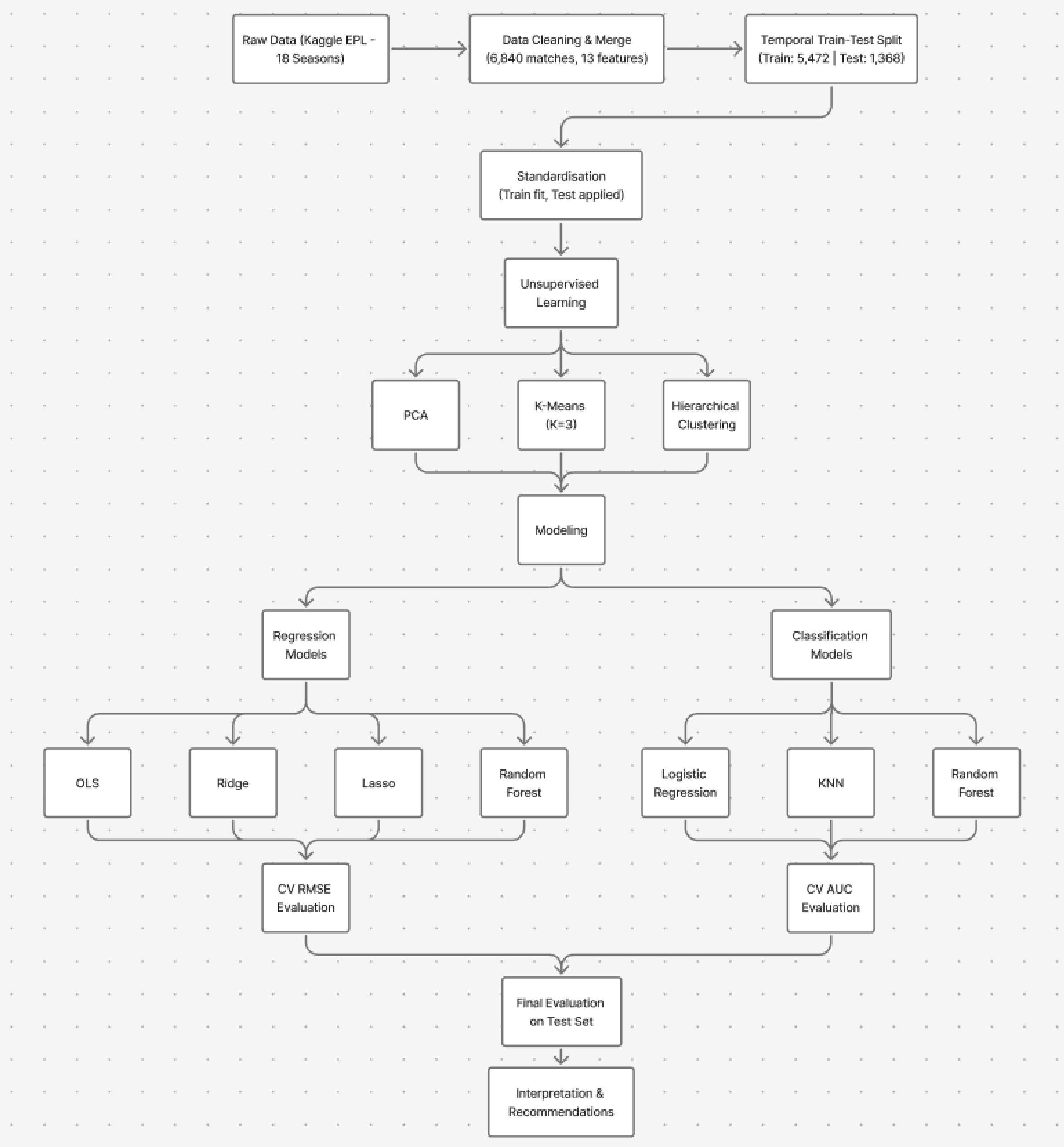
Appendices

Appendix A: Methodology Flowchart



Appendix B: Key R Code Snippets

The complete annotated R script is submitted separately as 'ST3189_230573190.R'. Key patterns are illustrated below.

B1. Temporal Split & Scaling Pipeline

```
train_idx <- 1:floor(0.8 * nrow(data))     # 5,472 rows (no leakage)
test_idx  <- (floor(0.8*nrow(data))+1):nrow(data)
glrn_ridge$id <- "scale.regr.ridge"        # unique ID for benchmark()
pipe <- po("scale") %>>% po("learner", lr_ridge)
glrn_ridge <- GraphLearner$new(pipe)
```

B2. K-means with Elbow Method

```
wss <- sapply(1:8, function(k) {
  kmeans(X_scaled[train_idx, ], centers=k, nstart=25)$tot.withinss
})
km3 <- kmeans(X_scaled[train_idx, ], centers=3, nstart=25)
```

B3. Classification Benchmark (5-fold CV, all models)

```
ll_knn <- lrn("classif.kknn", k=15, predict_type="prob")  # prob for ROC
bmr_cls <- benchmark(benchmark_grid(
  tasks=task_cls, learners=list(glrn_logit,glrn_lr,glrn_ll,glrn_knn,glrn_rfc),
  resamplings=rsmp("cv", folds=5)))
autoplot(bmr_cls, type="roc")
```

Appendix C: Full Numerical Results

C1. PCA Explained Variance

PC	Variance (%)	Cumulative (%)	Interpretation
PC1	31.6	31.6	Match Competitiveness
PC2	24.1	55.7	Match Quality
PC3	9.0	64.7	Mixed
PC4	7.1	71.9	Mixed
PC5	6.8	78.6	Mixed

C2. Logistic Ridge Coefficients (sorted by |coefficient|)

Feature	Coefficient	Direction
HTGD	+0.324	+ve: home strength ↑ win prob.
DiffFormPts	−0.279	−ve: away form ↓ win prob.
ATGD	−0.262	−ve: away strength ↓ win prob.
ATFormPts	−0.237	−ve: away form ↓ win prob.
HTFormPts	+0.235	+ve: home form ↑ win prob.
DiffPts	+0.158	+ve: points gap ↑ win prob.
ATP	−0.124	−ve: away quality ↓ win prob.
HTP	+0.076	+ve: home quality ↑ win prob.
Streak vars.	~0.000	Negligible after controlling form

Note: All figures above are generated directly by the submitted R script (ST3189_230573190.R) and reproduced here from the same underlying data for completeness.